

An exponential obstruction for rational equal-weight interval quadrature

Research note; proof candidate for independent verification

30 September 2026

Historical attribution. The parity polynomial $\sum_{j=1}^r (-2)^{j-1} \binom{x}{j}$ used below is classical: Mit'kin [1, Lemma 7 and following remark] credits it to Hua (1940), and records related Hilbert–Kamke applications by Arkhipov. The present note studies its integral after dilation and the consequence for rational equal-weight interval quadrature; no novelty claim is made for the polynomial itself.

Theorem 1. *Let $x_1, \dots, x_K$ be rational numbers (not necessarily in $[0, 1]$) with*

$$\frac{1}{K} \sum_{i=1}^K x_i^j = \frac{1}{j+1} \quad (0 \leq j \leq r).$$

Then $K \geq 2^{\lfloor r/2 \rfloor}$. More precisely:

1. *If every x_i belongs to $\mathbb{Z}_2$, then $2^r \mid K$.*
2. *Otherwise, after the normalization below, the nonzero number of odd nodes is divisible by $2^{\min(r, v_2(K) + \lfloor r/2 \rfloor)}$.*

Repeated nodes are allowed.

Status. The following is a complete proposed proof, with its coefficient identities also checked by exact rational arithmetic in the accompanying script. It is not a proof of the per-die exponential conjecture: the insertion reduction in the manuscript is only one-way. If valid, it rules out the proposed subexponential rational-quadrature route to counterexamples.

Define $G(z) = z/\log(1+z)$ as a formal series over $\mathbb{Q}$, with constant term one. The coefficient $[z^j]G(z)$ is $\int_0^1 \binom{x}{j} dx$, as follows by integrating the formal identity $(1+z)^x = \sum_{j \geq 0} \binom{x}{j} z^j$.

Put

$$D(t) = \frac{-\log(1-2t)}{2t} = \sum_{j \geq 0} \frac{2^j}{j+1} t^j.$$

Its coefficients are in $\mathbb{Z}_2$, and $D(t) \equiv 1+t \pmod{2}$: the terms $j=0,1$ are units, whereas $j - v_2(j+1) \geq 1$ for $j \geq 2$. Consequently $G(-2t) = D(t)^{-1} \equiv (1+t)^{-1} \pmod{2}$, so every coefficient of $G(-2t)$ is odd. In particular,

$$v_2 \left(\int_0^1 \binom{x}{j} dx \right) = -j. \quad (1)$$

If every $x_i \in \mathbb{Z}_2$, then $\binom{x_i}{r} \in \mathbb{Z}_2$. Exactness and (1) give $v_2(K) \geq r$, proving the first case.

For the other case, consider the degree- r polynomial

$$H_r(y) = \sum_{j=1}^r (-2)^{j-1} \binom{y}{j}.$$

For every $y \in \mathbb{Z}_2$,

$$H_r(y) \equiv \mathbf{1}_{y \notin 2\mathbb{Z}_2} \pmod{2^r}. \quad (2)$$

Indeed, for a nonnegative integer y , the binomial expansion of $(-1)^y$ gives this congruence after discarding terms of index greater than r . Continuity and density of the nonnegative integers extend it to $\mathbb{Z}_2$.

Lemma 2. *For every integer $a \geq 1$ and every integer $r \geq 1$,*

$$v_2 \left(\int_0^1 H_r(2^a x) dx \right) \geq \lfloor r/2 \rfloor.$$

Proof. Write $G(z) = \sum_{j \geq 0} g_j z^j$. We proved above that $v_2(g_j) = -j$. For an integer $N \geq 1$, coefficientwise integration gives

$$F_N(z) := \int_0^1 (1+z)^{Nx} dx = \frac{(1+z)^N - 1}{N \log(1+z)} = G((1+z)^N - 1).$$

The last identity follows from the formal identity $\log((1+z)^N) = N \log(1+z)$.

Suppose N is even and set $W_N(t) = (1-2t)^N - 1$. Its coefficient of degree $k \geq 1$ is $(-2)^k \binom{N}{k}$. For $k=1$ this has valuation at least 2; for $k \geq 2$ its valuation is at least $k \geq \lceil k/2 \rceil + 1$. Hence every coefficient of $W_N(t)^j$ in

degree k has valuation at least $\lceil k/2 \rceil + j$. Multiplying by g_j and summing proves that

$$F_N(-2t) = G(W_N(t)) = \sum_{k \geq 0} A_k t^k, \quad A_0 = 1, \quad v_2(A_k) \geq \lceil k/2 \rceil \quad (k \geq 1).$$

All these series converge on $\mathbb{Z}_2$. More precisely, $W_N(t) \in 4\mathbb{Z}_2$ for $t \in \mathbb{Z}_2$, so the valuation of $g_j W_N(t)^j$ is at least j , uniformly in t . Evaluation at $t = 1$ therefore commutes with the composition, and

$$\sum_{k \geq 0} A_k = G(W_N(1)) = G(0) = 1.$$

Apply this with $N = 2^a$. Finally,

$$\int_0^1 H_r(2^a x) dx = -\frac{1}{2} \sum_{k=1}^r A_k = \frac{1}{2} \sum_{k=r+1}^{\infty} A_k.$$

Its valuation is at least $\lceil (r+1)/2 \rceil - 1 = \lfloor r/2 \rfloor$. □

Conclusion of Theorem 1. Suppose some x_i is not 2-adically integral. Choose the smallest positive integer a such that $y_i = 2^a x_i \in \mathbb{Z}_2$ for every i . At least one y_i is odd. Let q be the number of odd y_i , so $1 \leq q \leq K$. Exactness, (2), and Lemma 2 yield

$$q \equiv \sum_{i=1}^K H_r(y_i) = K \int_0^1 H_r(2^a x) dx \pmod{2^r}.$$

In particular, $2^{\lfloor r/2 \rfloor} \mid q$, proving $K \geq 2^{\lfloor r/2 \rfloor}$. The displayed congruence also gives the more precise divisibility in item (2). □

References

- [1] D. A. Mit'kin, *An estimate for the number of terms in the Hilbert–Kamke problem*, Math. USSR Sbornik **57** (1987), no. 2, 561–590, Lemma 7 and its following remark, primary journal record and full text.